

Panel Data, Causal Inference and Difference-in-Differences

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Lecture Plan

- ▶ Part I: Introduction to Panel Data
 - ▶ Types of data structures
 - ▶ Unobserved heterogeneity and the unobserved effects model
 - ▶ First differencing and the fixed effects (within) estimator
 - ▶ Fixed effects vs. random effects
- ▶ Part II: Introduction to Causal Inference
 - ▶ Correlation vs. causation
 - ▶ The potential outcomes framework
 - ▶ Treatment effects, selection bias, and randomized experiments
- ▶ Part III: Difference-in-Differences (DiD)
 - ▶ The 2×2 design and the DiD estimator
 - ▶ Regression formulation and the parallel trends assumption
 - ▶ Application and extensions

Part I: Introduction to Panel Data

Part I

Introduction to Panel Data

Types of Data Structures

- ▶ **Cross-sectional data:** many units ($i = 1, \dots, n$) observed at a single point in time.

$$y_i = \beta_0 + \beta_1 x_i + u_i$$

- ▶ **Time series data:** a single unit observed over many periods ($t = 1, \dots, T$). This was the focus of most of this course.

$$y_t = \beta_0 + \beta_1 x_t + u_t$$

- ▶ **Pooled cross sections:** independent cross-sectional samples drawn at different points in time (different units each period).
- ▶ **Panel (longitudinal) data:** the *same* units i are observed over *multiple* periods t .

$$y_{it} = \beta_0 + \beta_1 x_{it} + u_{it}, \quad i = 1, \dots, n; \quad t = 1, \dots, T$$

What Does Panel Data Look Like?

- ▶ Each observation is identified by two indices: a cross-sectional unit i and a time period t .
- ▶ Example: data on n countries (or firms, individuals, cities) followed over T years.

unit (i)	year (t)	y_{it}	x_{it}
1	2018	$y_{1,2018}$	$x_{1,2018}$
1	2019	$y_{1,2019}$	$x_{1,2019}$
1	2020	$y_{1,2020}$	$x_{1,2020}$
2	2018	$y_{2,2018}$	$x_{2,2018}$
2	2019	$y_{2,2019}$	$x_{2,2019}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$

- ▶ A **balanced** panel has all T periods for every unit; otherwise it is **unbalanced**.

Why Use Panel Data?

- ▶ The key advantage: panel data lets us control for **unobserved, time-invariant** characteristics of the units.
- ▶ Many important variables are hard or impossible to measure: ability, motivation, managerial quality, geography, culture, institutions, “business climate”, etc.
- ▶ If these omitted factors are correlated with our regressors, OLS on a single cross section is biased (omitted variable bias).
- ▶ Idea: if such a factor does not change over time for a given unit, we can **eliminate** it by exploiting the repeated observations on the same unit.
- ▶ This is the panel-data route to a more credible *causal* interpretation.

The Unobserved Effects Model

- ▶ Consider the model

$$y_{it} = \beta_1 x_{it} + a_i + u_{it}, \quad t = 1, \dots, T$$

(we suppress the intercept / time dummies for now).

- ▶ a_i is the **unobserved effect** (also called the *fixed effect*, *individual heterogeneity*, or *unobserved heterogeneity*). It is constant over time but differs across units.
- ▶ u_{it} is the **idiosyncratic error**: it varies across both i and t .
- ▶ The problem: a_i is part of the error and may be correlated with x_{it} ,

$$\text{Cov}(x_{it}, a_i) \neq 0,$$

so pooled OLS (treating $a_i + u_{it}$ as one composite error) is biased and inconsistent.

Two Periods: First Differencing

- ▶ Suppose $T = 2$ (periods $t = 1, 2$). Write the model with a time intercept:

$$y_{it} = \beta_0 + \delta_0 d2_t + \beta_1 x_{it} + a_i + u_{it}$$

where $d2_t = 1$ in period 2 and 0 in period 1.

- ▶ Subtract the period-1 equation from the period-2 equation for the same unit i :

$$\underbrace{(y_{i2} - y_{i1})}_{\Delta y_i} = \delta_0 + \beta_1 \underbrace{(x_{i2} - x_{i1})}_{\Delta x_i} + \underbrace{(u_{i2} - u_{i1})}_{\Delta u_i}$$

- ▶ The unobserved effect a_i **drops out!** We can then estimate

$$\Delta y_i = \delta_0 + \beta_1 \Delta x_i + \Delta u_i$$

by OLS. This is the **first-differenced (FD) estimator**.

First Differencing: Assumptions

- ▶ For the FD estimator to be unbiased / consistent we need the differenced regressor to be uncorrelated with the differenced error,

$$\text{Cov}(\Delta x_i, \Delta u_i) = 0,$$

which follows from the strict exogeneity condition $E(u_{it} | x_{i1}, x_{i2}, a_i) = 0$.

- ▶ Crucially, we no longer need a_i to be uncorrelated with x_{it} — a_i has been differenced away. This is the power of panel data.
- ▶ We do need Δx_i to have variation across units (some units must change their x). Time-constant regressors cannot be included — they difference out too.
- ▶ With $T > 2$ periods, FD is applied to successive differences; the within (fixed effects) estimator is the more common alternative.

The Fixed Effects (Within) Estimator

- ▶ An alternative way to remove a_i with $T \geq 2$ is the **within transformation**.
- ▶ Start from $y_{it} = \beta_1 x_{it} + a_i + u_{it}$. Average over time for each unit i :

$$\bar{y}_i = \beta_1 \bar{x}_i + a_i + \bar{u}_i, \quad \bar{y}_i = \frac{1}{T} \sum_{t=1}^T y_{it}$$

- ▶ Subtract the time-average (the “within” demeaning):

$$(y_{it} - \bar{y}_i) = \beta_1 (x_{it} - \bar{x}_i) + (u_{it} - \bar{u}_i)$$

- ▶ Again a_i is eliminated. OLS on the demeaned data gives the **fixed effects (FE) estimator** $\hat{\beta}_1^{FE}$.
- ▶ Equivalently, FE = OLS with a separate dummy variable for each unit (the *Least Squares Dummy Variable*, LSDV, approach).

Fixed Effects: Remarks

- ▶ When $T = 2$, FE and FD give numerically identical estimates. For $T > 2$ they differ (FE is more efficient if u_{it} is serially uncorrelated; FD is preferred if u_{it} is close to a random walk).
- ▶ FE allows arbitrary correlation between a_i and the regressors x_{it} — this is its great strength.
- ▶ **Cost**: effects of *time-invariant* variables (e.g. gender, race, distance to coast) cannot be estimated; they are absorbed by a_i .
- ▶ In practice we also include **time dummies** to control for aggregate shocks common to all units in a period.
- ▶ Standard errors should be **clustered** at the unit level to allow for serial correlation within units.

Random Effects

- ▶ The **random effects (RE)** model keeps a_i in the composite error but assumes it is *uncorrelated* with the regressors:

$$\text{Cov}(x_{it}, a_i) = 0 \quad \text{for all } t.$$

- ▶ Under this assumption pooled OLS is consistent but inefficient (the composite error $v_{it} = a_i + u_{it}$ is serially correlated). RE uses a (feasible) GLS transformation that quasi-demeans the data:

$$y_{it} - \theta \bar{y}_i, \quad 0 \leq \theta \leq 1.$$

- ▶ RE is more efficient *if* its key assumption $\text{Cov}(x_{it}, a_i) = 0$ holds, and it can estimate coefficients on time-invariant variables.
- ▶ But if a_i is correlated with x_{it} , RE is **biased** — exactly the case FE handles.

Fixed Effects vs. Random Effects

- ▶ Central question: is the unobserved effect a_i correlated with the regressors?
 - ▶ **Yes** $\Rightarrow$ use Fixed Effects (RE is biased).
 - ▶ **No** $\Rightarrow$ Random Effects is consistent and more efficient.
- ▶ The **Hausman test** compares the FE and RE estimates:

$$H_0 : \text{Cov}(x_{it}, a_i) = 0 \quad (\text{RE appropriate}),$$

$$H_1 : \text{Cov}(x_{it}, a_i) \neq 0 \quad (\text{use FE}).$$

A large difference between $\hat{\beta}^{FE}$ and $\hat{\beta}^{RE}$ leads to rejecting H_0 .

- ▶ In applied microeconomics, FE is usually the default for causal questions because it is robust to a leading source of endogeneity.

Panel Data: Summary

- ▶ Panel data follow the same units over time.
- ▶ By differencing or demeaning, we can remove time-invariant unobserved heterogeneity a_i that would otherwise bias OLS.
- ▶ First differencing (FD) and fixed effects (FE) are the two workhorse estimators; they coincide when $T = 2$.
- ▶ This idea — using *within-unit* variation over time to control for confounders — is the bridge to modern causal inference and to difference-in-differences.

Part II: Introduction to Causal Inference

Part II

Introduction to Causal Inference

Correlation Is Not Causation

- ▶ Most of econometrics estimates conditional expectations / correlations. But economic policy questions are usually **causal**: “what is the effect of X on Y ?”
 - ▶ Does an extra year of schooling *cause* higher wages?
 - ▶ Does a minimum wage increase *cause* job losses?
 - ▶ Does a job-training program *cause* higher employment?
- ▶ A regression coefficient is causal only under strong assumptions. Threats:
 - ▶ **Omitted variables / confounding** (the “opera and longevity” example revisited).
 - ▶ **Reverse causality / simultaneity**.
 - ▶ **Selection** into treatment.
- ▶ Causal inference is a set of *research designs* that make the “other things equal” (*ceteris paribus*) comparison credible.

A Motivating Puzzle: Do Hospitals Make People Healthier?

- ▶ A simple question (from Angrist and Pischke, *Mostly Harmless Econometrics*): does going to the hospital make people healthier?
- ▶ Compare the self-reported health of people who went to the hospital with those who did not (health on a 1–5 scale: 1 = excellent, ..., 5 = poor, so **higher means worse**):

Group	Sample Size	Mean health status	Std. Error
Hospital	7774	2.79	0.014
No Hospital	90049	2.07	0.003

- ▶ The difference is $2.79 - 2.07 = 0.72$, statistically significant. Taken at face value, this says hospitals make people **sicker**!
- ▶ Clearly something is wrong with this comparison. What?

The Potential Outcomes Framework

- ▶ Framework due to Neyman and Rubin (the *Rubin Causal Model*).
- ▶ Let $D_i \in \{0, 1\}$ be a binary **treatment** (1 = treated, 0 = control / untreated).
- ▶ Each unit i has **two potential outcomes**:

$$y_i(1) = \text{outcome if treated}, \quad y_i(0) = \text{outcome if not treated}.$$
- ▶ The **causal effect** of the treatment for unit i is

$$\tau_i = y_i(1) - y_i(0).$$

- ▶ We only ever observe one of the two:

$$y_i = D_i y_i(1) + (1 - D_i) y_i(0).$$

- ▶ Hospital example: for person i , $y_i(1)$ is their health *if* hospitalized and $y_i(0)$ is their health *if not*. We see a sick person's $y_i(1)$, but never the $y_i(0)$ for the *same* person at the *same* time.

The Fundamental Problem of Causal Inference

- ▶ For any single unit, one potential outcome is observed and the other is the unobserved **counterfactual**.
- ▶ We can never compute $\tau_i = y_i(1) - y_i(0)$ for an individual — this is the **fundamental problem of causal inference** (Holland, 1986).
- ▶ So we focus on **average** effects across a population:

$$\text{ATE} = E[y_i(1) - y_i(0)], \quad \text{ATT} = E[y_i(1) - y_i(0) \mid D_i = 1].$$

- ▶ ATE = average treatment effect; ATT = average treatment effect on the treated.
- ▶ Estimating these requires constructing a credible counterfactual for the treated group.

Selection Bias

- ▶ The naive estimator is the difference in observed means between treated and untreated:

$$E[y_i \mid D_i = 1] - E[y_i \mid D_i = 0].$$

- ▶ Decompose it:

$$\begin{aligned} & E[y_i \mid D_i = 1] - E[y_i \mid D_i = 0] = \\ & \underbrace{E[y_i(1) \mid D_i = 1] - E[y_i(0) \mid D_i = 1]}_{\text{ATT (causal)}} \\ & + \underbrace{E[y_i(0) \mid D_i = 1] - E[y_i(0) \mid D_i = 0]}_{\text{selection bias}} \end{aligned}$$

- ▶ **Selection bias**: treated and untreated units differ in their untreated potential outcome $y_i(0)$, i.e. they are not comparable to begin with.
- ▶ Example: people who enroll in training may have lower earnings prospects anyway; comparing them to non-enrollees understates the program's effect.

Back to the Hospital Puzzle

- ▶ Now the puzzle is easy to explain. The naive difference mixes two things:

$$\underbrace{0.72}_{\text{observed}} = \underbrace{E[y_i(1) - y_i(0) \mid D_i = 1]}_{\text{true effect of hospital (ATT)}} + \underbrace{E[y_i(0) \mid D_i = 1] - E[y_i(0) \mid D_i = 0]}_{\text{selection bias}}.$$

- ▶ **Selection bias here is large and positive:** people who go to the hospital are sicker to begin with, so even untreated they would have worse health, $E[y_i(0) \mid D_i = 1] > E[y_i(0) \mid D_i = 0]$.
- ▶ The hospital may truly *improve* health (a negative ATT on this scale), but the selection bias is so large it swamps and even reverses the sign of the observed difference.
- ▶ **Lesson:** comparing treated vs. untreated is *not* the same as the causal effect, unless the two groups are comparable in their $y_i(0)$.

Randomized Experiments: The Gold Standard

- ▶ If treatment D_i is assigned **randomly**, it is independent of potential outcomes:

$$\{y_i(0), y_i(1)\} \perp D_i.$$

- ▶ Then treated and control groups are comparable on average, so $E[y_i(0) \mid D_i = 1] = E[y_i(0) \mid D_i = 0]$ and **selection bias is zero**.
- ▶ The simple difference in means consistently estimates the ATE:

$$\widehat{ATE} = \bar{y}_{\text{treated}} - \bar{y}_{\text{control}}.$$

- ▶ Randomized controlled trials (RCTs) are the benchmark, but often infeasible, unethical, or expensive in economics.
- ▶ This motivates **quasi-experimental** (observational) designs that mimic randomization.

A Menu of Identification Strategies

- ▶ When we cannot randomize, we look for a source of variation in D_i that is “as good as random”:
 - ▶ **Randomized experiments (RCTs)** — assignment is literally random.
 - ▶ **Instrumental variables (IV)** — a variable affecting D but not y directly.
 - ▶ **Regression discontinuity (RD)** — treatment switches at a cutoff of a running variable.
 - ▶ **Difference-in-differences (DiD)** — compare changes over time between treated and control groups.
- ▶ These are often based on **natural experiments**: policy changes, reforms, or events that affect some units but not others.
- ▶ We now study DiD in detail — it connects directly to the panel-data ideas from Part I.

Part III: Difference-in-Differences

Part III

Difference-in-Differences (DiD)

The Basic Idea

- ▶ Setting: a **policy / treatment** affects one group (the **treatment group**) but not another (the **control group**), and we observe both groups **before** and **after** the policy.
- ▶ Naive comparison 1 — *before vs. after* for the treated group: contaminated by other things that changed over time (aggregate trends).
- ▶ Naive comparison 2 — *treated vs. control after* the policy: contaminated by pre-existing differences between the groups (selection).
- ▶ **DiD idea**: combine the two. Use the control group's *change* over time to estimate what *would have happened* to the treated group absent the policy (the counterfactual).
- ▶ Take the **difference of the two differences**.

The 2 × 2 DiD Design

- ▶ Two groups (Treatment T , Control C) and two periods (Before, After).
- ▶ Let $\bar{y}$ denote the group–period average of the outcome.

	Before	After	After – Before
Treatment	$\bar{y}_{T,B}$	$\bar{y}_{T,A}$	$\Delta_T = \bar{y}_{T,A} - \bar{y}_{T,B}$
Control	$\bar{y}_{C,B}$	$\bar{y}_{C,A}$	$\Delta_C = \bar{y}_{C,A} - \bar{y}_{C,B}$
	DiD:		$\hat{\tau} = \Delta_T - \Delta_C$

- ▶ The DiD estimator:

$$\hat{\tau}_{DiD} = (\bar{y}_{T,A} - \bar{y}_{T,B}) - (\bar{y}_{C,A} - \bar{y}_{C,B}).$$

- ▶ Time-constant differences between groups and common time trends both cancel out.

DiD as a Regression

- ▶ Let $T_i = 1$ for the treatment group, $\text{Post}_t = 1$ in the after period. Estimate

$$y_{it} = \beta_0 + \beta_1 T_i + \beta_2 \text{Post}_t + \beta_3 (T_i \times \text{Post}_t) + u_{it}.$$

- ▶ Interpreting the coefficients via the group–period means:

	Before	After
Control	β_0	$\beta_0 + \beta_2$
Treatment	$\beta_0 + \beta_1$	$\beta_0 + \beta_1 + \beta_2 + \beta_3$

- ▶ The **interaction coefficient** is the DiD estimate:

$$\hat{\beta}_3 = (\bar{y}_{T,A} - \bar{y}_{T,B}) - (\bar{y}_{C,A} - \bar{y}_{C,B}) = \hat{\tau}_{DiD}$$

- ▶ The regression form gives standard errors directly and lets us add covariates.

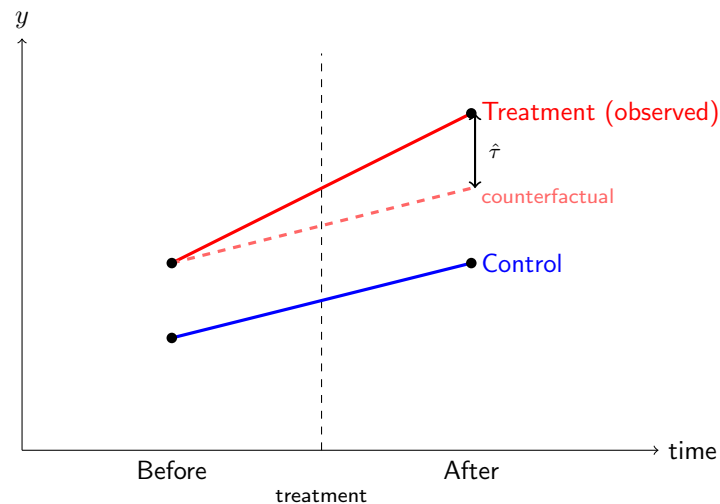
The Parallel Trends Assumption

- ▶ DiD identifies the causal effect **only if** the parallel trends assumption holds:
*In the absence of treatment, the treatment and control groups would have followed the **same trend** (same average change) in the outcome over time.*
- ▶ Formally, in potential-outcome terms:

$$E[y_{T,A}(0) - y_{T,B}(0)] = E[y_{C,A}(0) - y_{C,B}(0)].$$

- ▶ The levels of the two groups need *not* be equal — only their counterfactual *trends* must match.
- ▶ This assumption is **not directly testable** (the treated counterfactual is unobserved), but it is made more credible by inspecting **pre-treatment trends**.

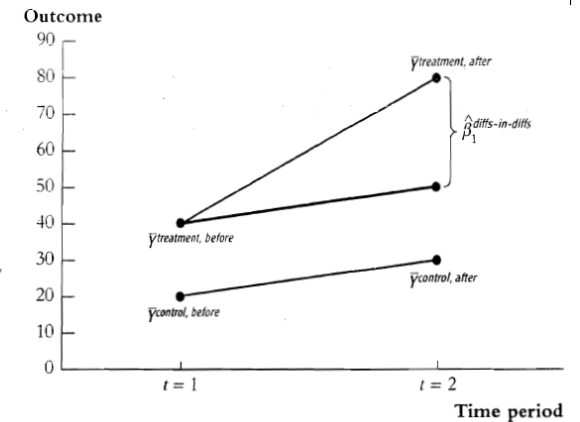
Parallel Trends: Illustration



- ▶ The DiD effect $\hat{\tau}$ is the gap between the treated group's *observed* outcome and its *counterfactual* (control trend applied to the treated group).

DiD Estimator in Two Periods: A Numerical Example

The post treatment difference between the treatment and control groups is $80 - 30 = 50$, but this overstates the treatment effect because before the treatment $\bar{y}$ was higher for the treatment than the control group by $40 - 20 = 20$. The differences-in-differences estimator is the difference between the final and initial gaps, so $\hat{\beta}_1^{\text{diffs-in-diffs}} = (80 - 30) - (40 - 20) = 50 - 20 = 30$. Equivalently, the differences-in-differences estimator is the average change for the treatment group minus the average change for the control group, that is,

$$\hat{\beta}_1^{\text{diffs-in-diffs}} = \Delta \bar{y}_{\text{treatment}} - \Delta \bar{y}_{\text{control}} = (80 - 40) - (30 - 20) = 30.$$


Source: Stock and Watson, *Introduction to Econometrics*, 3rd ed., ch. 13, p. 492.

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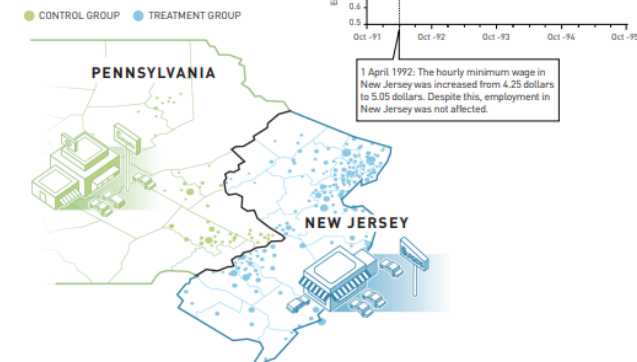
Classic Application: Minimum Wage and Employment

- ▶ Card and Krueger (1994), *Minimum Wages and Employment: A Case Study of the Fast-Food Industry in New Jersey and Pennsylvania*, *American Economic Review*, 84(4), 772–793.
- ▶ On April 1, 1992, New Jersey's minimum wage rose from \$4.25 to \$5.05 per hour. In neighboring Pennsylvania it stayed unchanged.
- ▶ Standard competitive theory predicts that a higher wage *reduces* labor demand, so employment should fall.
- ▶ They focus on the **fast-food industry**, which employs many low-skilled, minimum-wage workers and is easy to sample comparably across the two states.
- ▶ This is a classic **natural experiment**: a policy change affects one group (NJ) but not a comparable neighbor (PA).
- ▶ David Card was awarded the 2021 Nobel Memorial Prize in Economics, in part for this work.

The Card and Krueger Natural Experiment

The effect of increasing the minimum wage

Card and Krueger used a natural experiment to study how increasing the minimum wage affects employment. The researchers identified a treatment group (restaurants in New Jersey) and a control group (restaurants in eastern Pennsylvania) to measure the effect of increasing the minimum wage.



Source: The Nobel Committee for the Prize in Economic Sciences, nobelprize.org (2021).

Card and Krueger: Design and Data

- ▶ **Treatment group:** fast-food restaurants in New Jersey (minimum wage rose).
- ▶ **Control group:** fast-food restaurants in eastern Pennsylvania (no change).
- ▶ Why these states? They are neighbors, exposed to common macroeconomic shocks, and plausibly similar in many unobservable respects — making parallel trends credible.
- ▶ They surveyed the same restaurants in **two waves**: Wave I in Feb/March 1992 (*before*) and Wave II in Nov/Dec 1992 (*after*).
- ▶ Outcome: **full-time-equivalent (FTE) employment** = number of full-time workers (incl. managers) + 0.5 × number of part-time workers.
- ▶ DiD = (change in NJ employment) – (change in PA employment).

Card and Krueger (1994): Average Employment

TABLE 3—AVERAGE EMPLOYMENT PER STORE BEFORE AND AFTER THE RISE IN NEW JERSEY MINIMUM WAGE

Variable	Stores by state			Stores in New Jersey ^a			Differences within NJ ^b	
	PA (i)	NJ (ii)	Difference, NJ – PA (iii)	Wage = \$4.25 (iv)	Wage = \$4.26–\$4.99 (v)	Wage ≥ \$5.00 (vi)	Low–high (vii)	Midrange–high (viii)
1. FTE employment before, all available observations	23.33 (1.35)	20.44 (0.51)	–2.89 (1.44)	19.56 (0.77)	20.08 (0.84)	22.25 (1.14)	–2.69 (1.37)	–2.17 (1.41)
2. FTE employment after, all available observations	21.17 (0.94)	21.03 (0.52)	–0.14 (1.07)	20.88 (1.01)	20.96 (0.76)	20.21 (1.03)	0.67 (1.44)	0.75 (1.27)
3. Change in mean FTE employment	–2.16 (1.25)	0.59 (0.54)	2.76 (1.36)	1.32 (0.95)	0.87 (0.84)	–2.04 (1.14)	3.36 (1.48)	2.91 (1.41)
4. Change in mean FTE employment, balanced sample of stores ^c	–2.28 (1.25)	0.47 (0.48)	2.75 (1.34)	1.21 (0.82)	0.71 (0.69)	–2.16 (1.01)	3.36 (1.30)	2.87 (1.22)
5. Change in mean FTE employment, setting FTE at temporarily closed stores to 0 ^d	–2.28 (1.25)	0.23 (0.49)	2.51 (1.35)	0.90 (0.87)	0.49 (0.69)	–2.39 (1.02)	3.29 (1.34)	2.88 (1.23)

Notes: Standard errors are shown in parentheses. The sample consists of all stores with available data on employment. FTE (full-time-equivalent) employment counts each part-time worker as half a full-time worker. Employment at six closed stores is set to zero. Employment at four temporarily closed stores is treated as missing.

Source: Card and Krueger (1994), Table 3, p. 780. Standard errors in parentheses.

Card and Krueger: The DiD Estimate

- ▶ Average FTE employment per store (Table 3, columns (i)–(ii)):

	PA (control)	NJ (treatment)	NJ – PA
Before (Wave I)	23.33	20.44	–2.89
After (Wave II)	21.17	21.03	–0.14
Change (Δ)	–2.16	+0.59	

- ▶ The DiD estimate:

$$\hat{\tau}_{DiD} = \Delta \bar{y}_{NJ} - \Delta \bar{y}_{PA} = (0.59) - (-2.16) = +2.75 \text{ FTE workers.}$$

- ▶ Despite the wage increase, employment in NJ **did not fall** — if anything, it rose slightly relative to PA.
- ▶ This famous result challenged the simple competitive model of the labor market and helped launch the modern “credibility revolution” in empirical economics.

A Second Example: A Garbage Incinerator and House Prices

- ▶ Wooldridge's Examples 13.3–13.4 (based on Kiel and McClain, 1995): did a new garbage incinerator lower the price of nearby houses?
- ▶ A natural experiment in North Andover, Massachusetts: rumors about the incinerator began in **1978**, and it started operating in **1981**.
- ▶ Data set `kielmc`: **repeated cross sections** of house sales — 179 sales in **1978** (before) and 142 in **1981** (after), 321 in total.
- ▶ **Treatment group**: houses *near* the incinerator site (`nearinc=1`). **Control group**: houses *far* from it (`nearinc=0`).
- ▶ Outcome: `rprice`, the house price expressed in constant **1978 dollars** (so we strip out general inflation).
- ▶ Setup is exactly like Card–Krueger: two groups, two periods.

Incinerator: Step 1 — The 2×2 Table of Means

- Average real house price (*rprice*, 1978 dollars) by group and period:

	1978 (before)	1981 (after)	Change Δ
Far (control)	82,517	101,308	+18,790
Near (treatment)	63,693	70,619	+6,926
	DiD:		-11,864

- Near houses are cheaper *to begin with* ($\approx$ \$19,000 less in 1978) — a level difference that a simple “near vs. far” comparison would confound.
- DiD removes it by comparing *changes*:

$$\hat{\tau}_{DiD} = (+6,926) - (+18,790) = -\$11,864.$$

Incinerator: Step 2 — DiD as a Regression

- The same estimate comes from one OLS regression (Example 13.3):

$$rprice = \beta_0 + \beta_1 nearinc + \beta_2 y81 + \beta_3 (nearinc \times y81) + u$$

where $y81 = 1$ in 1981. The interaction β_3 **is** the DiD.

	Coefficient	Std. Error
Intercept	82,517	2,727
<i>nearinc</i>	-18,824	4,875
<i>y81</i>	18,790	4,050
<i>nearinc</i> $\times$ <i>y81</i> (DiD)	-11,864	7,457

- $\beta_3 = -11,864$ matches the table, but its *t*-ratio is only -1.59 — **not** statistically significant on its own.

Incinerator: Step 3 — Adding House Characteristics

- The simple DiD is imprecise because houses differ in many ways. Adding controls makes parallel trends more plausible (it need only hold *conditionally*) and sharpens the estimate (Example 13.4):

$$\log(price) = \beta_0 + \beta_1 nearinc + \beta_2 y81 + \beta_3 (nearinc \times y81) + \gamma^T x + u$$

where x adds age, age², rooms, baths, and (logs of) lot size, living area, and distance to the interstate.

- Now the DiD coefficient is

$$\hat{\beta}_3 = -0.132 \quad (\text{se } 0.052, \quad t = -2.53),$$

statistically significant.

- **Interpretation:** after the incinerator, houses near the site lost about **13%** of their value relative to comparable houses farther away.
- **Lesson:** controls can both sharpen the DiD estimate and strengthen the case for (conditional) parallel trends.

DiD and Panel Data

- DiD is closely related to the panel methods of Part I.
- With two periods, the interaction regression is exactly a first-differenced / fixed-effects style design: group fixed effects (T_i) absorb time-invariant group differences, and the time dummy ($Post_t$) absorbs common trends.
- With **panel data** (same units over time) we can include **unit fixed effects** and **time fixed effects**:

$$y_{it} = a_i + \lambda_t + \tau D_{it} + u_{it},$$

where $D_{it} = 1$ if unit i is treated in period t . This is the **two-way fixed effects (TWFE)** estimator.

- $\hat{\tau}$ generalizes the 2×2 DiD to many groups and many periods.

Extensions and Cautions

- ▶ **Event-study / pre-trends:** estimate effects period-by-period around the treatment date to check that pre-treatment effects are ≈ 0 (supporting parallel trends) and to trace dynamics.
- ▶ **Covariates:** add controls so that parallel trends needs to hold only conditionally.
- ▶ **Inference:** cluster standard errors at the group level (Bertrand, Duflo, Mullainathan, 2004, warn about serial correlation in DiD).
- ▶ **Staggered adoption:** when units are treated at *different* times, the simple TWFE estimator can be biased (recent literature: Goodman-Bacon; Callaway & Sant'Anna; de Chaisemartin & D'Haultfœuille). Newer “robust” DiD estimators address this.
- ▶ Threats: differential trends, anticipation effects, spillovers between groups, and compositional changes.

Course Wrap-Up

- ▶ **Panel data** let us control for unobserved, time-invariant heterogeneity via first differencing or fixed effects.
- ▶ **Causal inference** reframes empirical questions in terms of potential outcomes, treatment effects, and the central problem of constructing a credible counterfactual.
- ▶ **Difference-in-differences** is a transparent, widely used quasi-experimental design that uses a control group's trend as the counterfactual, under the parallel trends assumption.
- ▶ Together these tools move us from *correlation* toward credible *causal* statements — the goal of modern applied econometrics.

This concludes Econometrics II. Thank you!

References / Further Reading

- ▶ Wooldridge, J. *Introductory Econometrics: A Modern Approach* — Ch. 13 (pooled cross sections & panel data, DiD) and Ch. 14 (fixed and random effects).
- ▶ Angrist, J. & Pischke, J.-S. (2009). *Mostly Harmless Econometrics*.
- ▶ Card, D. & Krueger, A. (1994). “Minimum Wages and Employment.” *American Economic Review*.
- ▶ Cunningham, S. (2021). *Causal Inference: The Mixtape*.